

Zhantong Qiu

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INTRODUCTION

Computer architecture researcher focusing on computer system performance evaluation, simulation methodology, and hardware-software co-design for emerging applications. I build novel frameworks for fast, accurate, and portable computer system performance evaluation using sampled simulation. I tackle robotic applications on tiny robots and HPC workloads in datacenters through hardware-software co-design. I am a strong programmer and active contributor to production open-source simulators (e.g., gem5). I seek challenges in constrained systems and solve them with solutions that span multiple layers of the computer system stack.

EDUCATION

Master of Science in Computer Science <i>University of California, Davis Davis, California</i>	Fall 2023 – Spring 2026 (expected)
Visiting Student in Electrical and Computer Engineering <i>Cornell University Ithaca, New York</i>	Fall 2025 – Spring 2026
Bachelor of Science in Computer Science and Engineering <i>University of California, Davis Davis, California</i>	Fall 2020 – Spring 2023

PUBLICATIONS

- **Nugget: Portable Program Snippets⁴** (Accepted to HPCA 2026)

Authors: **Zhantong Qiu**, Mahyar Samani, Jason Lowe-Power.

Nugget introduces an LLVM IR-based sampling framework that turns long-running workloads into portable program snippets (“nuggets”) which can be analyzed once, then reused across binaries, ISAs, and microarchitectures. It replaces simulator-based interval analysis with compiler instrumentation on real hardware, cutting sampling overhead by up to hundreds× versus functional simulation, while keeping intervals binary-independent via an IR-level unit of work. Using large SPEC CPU2017, NPB, and LSMS workloads, the work shows how Nugget enables fast, hardware-based validation of sampling methods and helps diagnose simulator modeling errors in gem5’s Arm memory system.

- **Accelerating the Simulation of Parallel Workloads using Loop-Bounded Checkpoints (Under review at ACM TACO)**

Authors: Alen Sabu, **Zhantong Qiu**, Harish Patil, Changxi Liu, Wim Heirman, Jason Lowe-Power, Trevor E. Carlson.

Proposed *LoopPoint*, a synchronization-agnostic loop-based sampling methodology that enables fast, accurate simulation of multi-threaded workloads via loop-bounded checkpoints, achieving up to 801× speedup on SPEC CPU2017 with ~2.3% average runtime error, and introduced *ROIperf* for validating representative regions directly on silicon. This work extends the prior *LoopPoint* paper published at HPCA.

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⁴<https://arxiv.org/abs/2509.02873>

TALKS AND TUTORIALS

- [HPCA 2023 LoopPoint Tools: Sampled Simulation of Complex Multi-threaded Workloads using Sniper and gem5⁵](#)
- [ISCA 2023 gem5 Workshop⁶](#)
- [gem5 Bootcamp 2024⁷](#)

RESEARCH EXPERIENCE

Graduate Researcher

June 2023 – Present

DArchR Lab, UC Davis

Advisor: Jason Lowe-Power

- **Nugget (LLVM IR sampling)**
 - Designed and implemented a cross-platform, architecture-independent sampling framework at the LLVM IR level; supports rapid interval analysis, native execution for validation, and cross-ISA simulation.
 - Evaluated Nugget on real hardware platforms with diverse performance characteristics (e.g., Ampere Altra) and in the gem5 simulator to validate fidelity and portability.
 - Analyzed a diverse set of workloads (SPEC CPU2017, NPB, LSMS) to demonstrate robustness across CPU, HPC, and scientific applications.
- **Accelerating the Simulation of Parallel Workloads using Loop-Bounded Checkpoints**
 - Co-led methodology and implementation to validate LoopPoint samples on gem5; under journal review.
- **gem5 contributions**
 - Ongoing upstream contributions since 2022 to full-system sampling support and related simulation features in gem5; author of [50 commits⁸](#) (public history).

Visiting Student

August 2025 – Present

Computer Systems Laboratory, Cornell University

Advisor: Christopher Batten

- **Agile Robotic Hardware-Software Co-Design**
 - Developing a closed-loop, multi-robot evaluation framework integrating robotics simulators and architecture simulators; enables parallel per-robot simulation and decouples from a specific ISA/simulator to broaden studies (e.g., ISA extensions).
- **Accurate STM32G4 Board in gem5**
 - Building an accurate STM32G4 MCU model in gem5 (Arm M-profile). Implemented the [ART \(Adaptive Real-Time\) accelerator⁹](#) to improve Flash access latency; planning to upstream core M-profile components (e.g., NVIC).

⁵<https://looppoint.github.io/hpca2023/>

⁶<https://www.gem5.org/events/isca-2023>

⁷<https://gem5bootcamp.github.io/2024/>

⁸<https://github.com/gem5/gem5/commits?author=studyztp>

⁹<https://github.com/studyztp/gem5/tree/studyztp/ART-cache>

- **SimPoint**

- Implemented SimPoint support in the gem5 stdlib and ran SPEC CPU2006 experiments using SimPoint.

- **LoopPoint**

- Implemented LoopPoint support in system-emulation and full-system modes in gem5; added a generalized PC-execution counter to track occurrences of specific PCs.

COURSE PROJECTS

CXL Shared Memory Filesystem Optimization¹⁰

Investigated application needs in a CXL shared-memory system and improved the FAMFS framework with more efficient allocation/deallocation and zero-copy operations to avoid redundant copies.

Limitations of Disaggregated Memory and Innovations¹¹

Analyzed limitations of disaggregated memory and proposed a hardware-software co-designed page management approach.

CUDA Microbenchmarks¹²

Developed configurable microbenchmarks to measure shared-memory latency and memory-scaling behavior on GPUs.

RISC-V Operating System

Built a simple RISC-V OS on the UC Davis RISC-V Console Simulator for a course project.

TEACHING EXPERIENCE

WQ 2024 ECS 154B: Computer Architecture Teaching Assistant

Led weekly discussion sections and office hours; created assignment material on Chisel-based CPU model (DINO CPU).

ORGANIZATIONS

Founder of Computer Systems Seminar at UC Davis¹³

Organized weekly speaker series with 12+ talks from academia and industry (still on-going).

SKILLS

Programming Languages: C/C++, Python, Bash, CUDA, assembly, Latex

Hardware Description Languages: Chisel

System Evaluation Tools: gem5, QEMU

Profiling & Instrumentation: LLVM passes, DynamoRIO, Valgrind, Linux perf, PAPI

Other Linux Related Tools: cpuset, CRIU, cgroups, Docker

Compilers: GCC/G++, GFortran, Clang/LLVM

Languages: English, Cantonese, Mandarin

¹⁰<https://github.com/orgs/ECS-289D/repositories>

¹¹https://github.com/studyztpt/my-course-paper/blob/main/Limitations_of_Disaggregated_Memory_and_Innovations.pdf

¹²https://github.com/studyztpt/CUDA_microbenchmarks.git

¹³<https://arch.cs.ucdavis.edu/events/computer-system-seminar>